

Parliamentarians

Tim Sherratt

The [Parliamentary Handbook](#) provides a wealth of information on the make up of the Australian Parliament, including biographical details of all parliamentarians from 1901.

```
import pandas as pd
import altair as alt
from pandas import json_normalize
import requests
```

```
response = requests.get("https://handbookapi.aph.gov.au/api/individuals/")

df = json_normalize(response.json()["value"])
df.head()
```

	PHID	GivenName	MiddleNames	FamilyName	PreferredName	DisplayName
0	EZ5	Anthony	John	ABBOTT	(Tony)	ABBOTT, the Hon. Anthony (Tony)
1	JLJ	Charles	Lydiard Aubrey	ABBOTT		ABBOTT, the Hon. Charles Lydiar
2	JLL	Joseph	Palmer	ABBOTT		ABBOTT, the Hon. Joseph Palmer
3	JNT	Macartney		ABBOTT		ABBOTT, Macartney
4	JLM	Percy	Phipps	ABBOTT		ABBOTT, Percy Phipps, CMG

```
gender_total = df["Gender"].value_counts().to_frame().reset_index()
gender_house = df.loc[df["ElectedMemberNo"] > -1]["Gender"].value_counts().to_frame().reset_index()
gender_senate = df.loc[df["ElectedSenatorNo"] > -1]["Gender"].value_counts().to_frame().reset_index()
gender_total
```

	Gender	count
0	Male	1584
1	Female	295

```

base_total = alt.Chart(gender).encode(
    theta=alt.Theta("count").stack(True),
    color=alt.Color("Gender").legend(None)
).properties(width="container")

total_pie = base_total.mark_arc()
total_text = base_total.mark_text(radius=170, size=12).encode(text="Gender:N")
chart_total = alt.layer(total_pie, total_text, padding=20)

base_house = alt.Chart(gender_house).mark_arc().encode(
    theta=alt.Theta("count").stack(True),
    color=alt.Color("Gender").legend(None)
).properties(width="container")

house_pie = base_house.mark_arc(outerRadius=120)
house_text = base_house.mark_text(radius=140, size=12).encode(text="Gender:N")
chart_house = alt.layer(house_pie, house_text, padding=20)

base_senate = alt.Chart(gender_senate).mark_arc().encode(
    theta=alt.Theta("count").stack(True),
    color=alt.Color("Gender").legend(None)
).properties(width="container")

senate_pie = base_senate.mark_arc(outerRadius=120)
senate_text = base_senate.mark_text(radius=140, size=12).encode(text="Gender:N")
chart_senate = alt.layer(senate_pie, senate_text, padding=20)

display(chart_total, chart_house, chart_senate)

```

```

gender_house = df.loc[df["ElectedMemberNo"] > -1]["Gender"].value_counts().to_frame().reset_index()
gender_senate = df.loc[df["ElectedSenatorNo"] > -1]["Gender"].value_counts().to_frame().reset_index()

```

```
alt.LayerChart(...)
```

```
alt.LayerChart(...) (a) Both houses alt.LayerChart(...)
```

```
(b) House of Representatives
```

```
(c) Senate
```

Figure 1: Number of parliamentarians by gender

```
chart1 = alt.Chart(gender_house).mark_arc().encode(
    theta="count",
    color="Gender"
).properties(width="container")

chart2 = alt.Chart(gender_senate).mark_arc().encode(
    theta="count",
    color="Gender"
).properties(width="container")

display(chart1, chart2)
```

```
alt.Chart(...)
```

```
alt.Chart(...)
```

```
(a) Membership by gender
```

```
(b)
```

```
df_parliaments = df.copy()[["PHID", "GivenName", "MiddleNames", "FamilyName", "DateOfBirth", "Gender", "RepresentedParliaments"]]
```

```
df_parliaments = df_parliaments.explode("RepresentedParliaments", ignore_index=True)
```

```
df_parliaments
```

	PHID	GivenName	MiddleNames	FamilyName	DateOfBirth	Gender	RepresentedParliaments
0	EZ5	Anthony	John	ABBOTT	1957-11-04	Male	37
1	EZ5	Anthony	John	ABBOTT	1957-11-04	Male	38
2	EZ5	Anthony	John	ABBOTT	1957-11-04	Male	39
3	EZ5	Anthony	John	ABBOTT	1957-11-04	Male	40
4	EZ5	Anthony	John	ABBOTT	1957-11-04	Male	41

	PHID	GivenName	MiddleNames	FamilyName	DateOfBirth	Gender	RepresentedParliaments
...	...	...	...	...	...	...	...
8753	KC7	William	Austin	ZEAL	1830-12-05	Male	1
8754	KC7	William	Austin	ZEAL	1830-12-05	Male	2
8755	203092	Trent	Moir	ZIMMERMAN	1968-10-15	Male	44
8756	203092	Trent	Moir	ZIMMERMAN	1968-10-15	Male	45
8757	203092	Trent	Moir	ZIMMERMAN	1968-10-15	Male	46

```
parl_gender = df_parliaments.value_counts(["Gender", "RepresentedParliaments"]).to_frame().reset_index()
parl_gender
```

	Gender	RepresentedParliaments	count
0	Male	37	209
1	Male	34	208
2	Male	36	208
3	Male	35	206
4	Male	25	203
...	...	...	...
75	Female	18	4
76	Female	27	4
77	Female	20	4
78	Female	28	2
79	Female	17	2

```
alt.Chart(parl_gender).mark_bar().encode(
    x="RepresentedParliaments:O",
    y="count:Q",
    color="Gender"
).properties(width="container")
```

```
alt.Chart(...)
```

```
df_parliaments.loc[df_parliaments["ElectedMemberNo"] != -1]
```

	PHID	GivenName	MiddleNames	FamilyName	DateOfBirth	Gender	RepresentedParliamen
0	EZ5	Anthony	John	ABBOTT	1957-11-04	Male	[37, 38, 39, 40, 41, 42,
1	JLJ	Charles	Lydiard Aubrey	ABBOTT	1886-01-04	Male	[10, 11, 13, 14]
2	JLL	Joseph	Palmer	ABBOTT	1891-10-18	Male	[16, 17, 18]
4	JLM	Percy	Phipps	ABBOTT	1869-05-14	Male	[5, 6, 7, 10, 11]
6	316915	Basem		ABDO		Male	[48]
...	...	...	...	...	...	...	...
1872	201906	Terry	James	YOUNG	1968-02-20	Male	[46, 47, 48]
1873	84H	Christian	John	ZAHRA	1973-04-08	Male	[39, 40]
1875	ZK6	Paul	John	ZAMMIT	1941-04-28	Male	[38]
1876	HWB	Antonio		ZAPPIA	1952-06-13	Male	[42, 43, 44, 45, 46, 47,
1878	203092	Trent	Moir	ZIMMERMAN	1968-10-15	Male	[44, 45, 46]

```
df_parliaments.loc[df_parliaments["ElectedSenatorNo"] != -1]
```

	PHID	GivenName	MiddleNames	FamilyName	DateOfBirth	Gender	RepresentedParliaments
3	JNT	Macartney		ABBOTT	1877-07-03	Male	[14, 15, 16]
4	JLM	Percy	Phipps	ABBOTT	1869-05-14	Male	[5, 6, 7, 10, 11]
5	JYU	Richard	Hartley Smith	ABBOTT	1859	Male	[11]
8	N26	Eric		ABETZ	1958-01-25	Male	[37, 38, 39, 40, 41, 42, 43,
10	E4Q	Judith	Anne	ADAMS	1943-04-11	Female	[41, 42, 43]
...	...	...	...	...	...	...	...
1864	KBW	Reginald	Charles	WRIGHT	1905-07-10	Male	[19, 20, 21, 22, 23, 24, 25,
1867	8IV	Nicholas		XENOPHON	1959-01-29	Male	[42, 43, 44, 45]
1870	KBY	Harold	William	YOUNG	1923-06-30	Male	[26, 27, 28, 29, 30, 31, 32]
1874	FE4	Alice	Olive	ZAKHAROV	1929-03-19	Female	[33, 34, 35, 36, 37]
1877	KC7	William	Austin	ZEAL	1830-12-05	Male	[1, 2]